

AI-Powered Task Management Portal

1. Assumptions

- Each user can access and manage only their own tasks.
- A user must register and log in before accessing the task management features.
- JWT is used for authentication and protected API requests.
- The task contains a title, description, priority, due date, status and estimated completion time.
- AI-generated task details are suggestions. The user can edit the generated values before saving the task.
- If the AI service is unavailable or fails, the user can enter the task details manually.
- Database credentials, JWT secret and Gemini API key are provided through environment variables and are not stored in the GitHub repository.

2. AI Workflow

The application uses the Gemini API to help the user create task details from a task title.

1. The user enters a task title in the React frontend.
2. The user clicks the Generate with AI button.
3. React sends the task title to the Spring Boot AI endpoint.
4. The backend creates a prompt and sends it to the Gemini API.
5. Gemini returns the generated task information.
6. The backend processes the AI response and sends the result back to React.
7. The frontend fills the task description, priority and estimated hours automatically.
8. The user can review or edit the generated values and then save the task.
9. If the AI request fails, fallback/default values are used so the task can still be created manually.

AI Flow: Task Title → React → Spring Boot → Gemini API → Backend Response → React Form

3. Blockchain Implementation

Blockchain was not attempted in this project. The project focuses on task management, JWT authentication, database operations and AI integration.

4. Project Summary

The project is a full-stack task management portal built with React and Spring Boot. It provides user authentication, task CRUD operations, task status management and Gemini AI integration for generating useful task details.

5. Links

- GitHub - <https://github.com/sid581/AI-Powered-Task-Management-Portal>
- Demo Video - https://drive.google.com/file/d/1DJU7bmg2vWvj-pdrylg1PzU8z_olZp3u/view?usp=sharing